Aditya Gowari

M.Sc. Scientific Computing | HPC & GPU Systems
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LinkedIn: linkedin.com/in/aditya-gowari | GitHub: github.com/patzer-adi | LeetCode: leetcode.com/u/patzer_adi_22 (101+ solved)

SUMMARY

M.Sc. Scientific Computing student with a focus on HPC, GPU computing, and computational cryptography. Comfortable with CUDA/MPI parallel system design, numerical algorithm development, and C/C++, and currently building a stronger foundation in statistical methods and scientific Python through ongoing research in observational cosmology.

TECHNICAL SKILLS

Languages: C/C++, Python
HPC / Parallel Computing: CUDA, MPI, OpenMP, Multithreading, Kernel & GPU Memory Optimization, PBS
Libraries / Scripting: NTL, GMP, NumPy, SciPy, SymPy, Matplotlib, SageMath, Shell Scripting
Dev Tools: Git, CMake, GTest, Valgrind, SSH, LaTeX, Jupyter Notebook, VSCode
Creative Tools: DaVinci Resolve, Canva
Systems: Linux (Arch, Ubuntu, Debian), macOS, Windows

RESEARCH EXPERIENCE

Vacation Students Programme (VSP) Research Intern June 2026 -- Present
Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune *Supervisor: Prof. Aseem Paranjape*
Project: The Galaxy--Halo Connection for Mock Catalogue Generation
Tech Stack: Python, Jupyter Notebook, PBS, SSH, NumPy, SciPy, SymPy, Matplotlib, Shell Scripting

- Working with Halo Occupation Distribution (HOD) models to populate dark matter halos from cosmological simulations with galaxies, generating mock galaxy catalogues and studying their large-scale clustering statistics for applications to next-generation (Stage-4) cosmological surveys.
- Building the statistical foundation needed for the pipeline through guided reading and weekly discussions with Prof. Paranjape.

PROJECTS

1) Efficient Algorithms for the Zero Minor Problem Aug 2025 -- Present
GitHub: *github.com/patzer-adi/zero_minor_problem*
Research project on detecting zero-determinant minors in $n \times n$ ECDLP kernel matrices --- a structural problem at the intersection of linear algebra and computational cryptography. Team of 4.
Tech Stack: C++, NTL, GMP, CUDA, MPI, CMake, Python, NumPy, Matplotlib, PBS, SSH, Linux
Responsibilities:

- Built a CUDA APM kernel where each thread extracts a $k \times k$ submatrix, runs modular Gaussian elimination (det mod p), and signals zero results via `atomicOr`; achieved an **81% detection rate**, scaling to encryption level 40 on NVIDIA V100 (PARAM Brahma, **IISER Pune**), RTX 2080 Ti, and 750 Ti over **150+ hours** of compute (including a single ∼**70-hour** run).
- Generated **1000+** structurally valid ECDLP matrices across a **48-core MPI cluster** (**300+ hours**); results are narrowing in on the minimum deviation level at which zero APM minors appear consistently --- a key structural threshold for the team's ongoing analysis.
- Patched the `lasvegas-ecdlp` C++/MPI codebase to generate matrices at scaled offset 0.2 --- fixed offset type and `numberOfBits` computation, enabling the antioffset dataset --- then helped analyze and visualize results using Python, NumPy, and Matplotlib.

2) Numerical Computing Library in C++ Jan 2026 -- Present
GitHub: *github.com/patzer-adi/Numerical_Computing*
C++ numerical methods library with runtime GPU/CPU dispatch --- algorithms execute on CUDA when a GPU is available, falling back to CPU automatically.
Tech Stack: C++, CUDA, CMake, GTest, Valgrind, Linux (Arch, Debian)
Responsibilities:

- Implemented Gaussian Elimination, Gauss-Jordan, LU Decomposition (Doolittle, Crout, Cholesky), Gauss-Jacobi, and Gauss-Seidel; validated on matrices up to **225×225** across CPU and CUDA.
- Built a `Complex` class with full arithmetic and root-finding solvers: Newton-Raphson, Fixed-Point Iteration, and Bisection Method.

EDUCATION

Master of Science in Scientific Computing 2025 -- 2027
SCMS, Savitribai Phule Pune University, Pune *(Pursuing)*

Bachelor of Science in Computer Science 2022 -- 2025
MES Abasaheb Garware College, Savitribai Phule Pune University, Pune **CGPA: 9.03/10**

EXTRACURRICULAR ACTIVITIES

- Edited/produced a podcast featured on the School of Scientific Computing YouTube channel; co-hosted a 100-guest alumni meet (Aug 2025); organized and competed in a chess tournament.